Exercise Sheet 7: Eigenvalues and Eigenvectors

December 2024

In Exercises 1-2, find (a) the characteristic equation of A, (b) the (real) eigenvalue of A, and (c) a basis for the eigenspace corresponding to each eigenvalue.

1. $A = \begin{bmatrix} 2 & 1 \\ 5 & -2 \end{bmatrix}$.

2. $A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 1 & 0 & -2 \end{bmatrix}$.

In Exercises 3-6, determine whether A is diagonalizable. If it is, find a nonsingular matrix P such that $P^{-1}AP$ is diagonal.

3. $A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -1 & 1 & 0 \end{bmatrix}$.

4. $A = \begin{bmatrix} -2 & -1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$.

5. $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 3 & 0 \\ 4 & -2 & 5 \end{bmatrix}$.

6. $A = \begin{bmatrix} 1 & 3 \\ -1 & 5 \end{bmatrix}$.

7. Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3, T(x, y, z) = (-2x + 2y - 3z, 2x + y - 6z, -x - 2y)$. Find a basis B for $\mathbb{R}^3$ such that the matrix of T relative to B is diagonal.

8. Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}$. Find A^{2024}.

. In Exercises 9-10, find an orthogonal matrix P that diagonalizes A.

9. $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$.

10. $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.